NATIONAL CADET CORPS
CADET MANAGEMENT SYSTEM
PROJECT REPORT

Project Title	NCC Cadet Management System
Version	1.0.0
Release Status	Final Release
Report Date	April 2026
Developed By	Swarda Avhad
Live URL	https://ncc-1-maharashtra.vercel.app/
Repository	github.com/swardaavhad-afk/NCC-1-Maharashtra
Platform	Node.js + Vanilla JS SPA + PostgreSQL (Supabase)
Deployment	Vercel Edge Serverless + Global CDN

Full Stack Web Development Project | Academic Year 2025–2026

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EXECUTIVE SUMMARY

The NCC Cadet Management System is an enterprise-grade, full-stack web application engineered to strategically digitize, automate, and centralize the operational and administrative workflows of the National Cadet Corps (NCC). Historically, NCC units relied heavily on decentralized, paper-based workflows to manage cadet data, track camp enrollments, monitor attendance, and log achievements. This traditional approach proved prone to data redundancy, operational inefficiencies, and security vulnerabilities.

This platform delivers a robust, role-based cloud solution that bridges the logistical gap between Administrative Officers and participating cadets. Built on a modern technology stack featuring a high-performance Node.js REST API, an interactive Vanilla JavaScript Single Page Application (SPA), and a managed PostgreSQL relational database, the system ensures seamless data integrity, scalable performance, and real-time operational governance.

Key Project Metrics at a Glance

METRIC	DETAIL	METRIC	DETAIL
Project Version	1.0.0 (Final Release)	**Status**	Production Deployed
Release Date	April 2026	**Architecture**	Decoupled Client-Server
Frontend Runtime	Vanilla JS (ES6+) SPA	**Backend Runtime**	Node.js v18+ / Express.js
Database	PostgreSQL via Supabase	**Authentication**	Supabase Auth / JWT
Deployment	Vercel Edge Serverless	**Primary Users**	Officers & Cadets

1. INTRODUCTION

1.1 Project Background

The National Cadet Corps (NCC) conducts nationwide youth development programs focusing on discipline, leadership, and structured training. Executing these programs involves tracking detailed profiles, managing logistical requirements for training camps, and monitoring the day-to-day engagement of thousands of cadets across three wings: Army, Navy, and Air Force.

The NCC Cadet Management System transitions these widespread organizational elements into a cohesive, cloud-based portal — accessible anywhere, at any time, with zero paper overhead.

1.2 Project Objectives

- **Digitization:** Migrate physical ledgers, cadet service books, and attendance sheets into a structured, queryable PostgreSQL database.
- **Role-Based Automation:** Enforce a secure operational hierarchy by isolating administrative capabilities from cadet self-service functionalities.
- **Resource Accessibility:** Provide a centralized hub for distributing digital study materials, camp itineraries, and operational notices.
- **Data Visualization:** Give administrators statistical overviews of corps performance through KPI-driven executive dashboards.

- **Scalability:** Build an infrastructure ready to accommodate future growth — multi-unit support, biometric integration, and predictive analytics.

2. PROBLEM STATEMENT & SCOPE

2.1 Limitations of the Existing System

The current cadet record management process is highly fragmented. Enrollment forms are distributed manually, camp applications rely on physical signatures, and attendance is recorded in physical registers. This results in the following critical operational bottlenecks:

- **High Margin of Error:** Manual entry mistakes, lost documents, and mismatched cadet ID numbers create data integrity issues.
- **No Real-Time Analytics:** Administrators cannot instantly generate performance reports or visualize corps-wide statistics.
- **Inefficient Communication:** Announcements regarding camps and study materials are delayed due to decentralized communication channels.
- **Data Redundancy:** Same cadet information maintained across multiple physical registers leads to version conflicts.
- **Security Vulnerabilities:** Paper records offer no access control — anyone with physical access can view or alter records.

2.2 The Proposed Digital Solution

The NCC Cadet Management System resolves all aforementioned bottlenecks by introducing a fully digital, role-protected operational ecosystem:

- A streamlined e-registration pipeline enforcing strict NCC enrollment number format guidelines.
- A live, role-guarded digital ecosystem tracking cadet progression from onboarding to achievement certification.
- An interactive SPA interface accessible across all desktop environments.
- A 'Single Source of Truth' database eliminating all data redundancy and inconsistency.
- JWT-based access control ensuring zero privilege escalation between user roles.

3. SYSTEM ARCHITECTURE & TECHNOLOGY STACK

3.1 Architectural Paradigm

The application employs a decoupled Client-Server Architecture utilizing a RESTful service paradigm. This architecture guarantees a distinct separation of concerns between UI presentation and server-side business logic processing.

Architectural Layers

LAYER	COMPONENTS	DESCRIPTION
CLIENT LAYER	Vanilla JS SPA (ES6+) | HTML5 | CSS3 Custom Grid	Single Page Application rendered in the browser. DOM manipulation via event-driven JS modules. No framework overhead — pure ES6 modules communicate with the backend via Fetch API calls. Local sessionStorage caches JWT tokens and user role for offline-tolerant UX.

API GATEWAY LAYER	Express.js REST API | Node.js v18+ | auth.js Middleware	Stateless HTTP server exposing RESTful endpoints grouped by domain (/auth, /cadets, /camps, /attendance, /achievements, /materials). Every protected route passes through auth.js which validates the JWT Bearer token and enforces RBAC before delegating to controller functions.
BUSINESS LOGIC LAYER	Controllers: authController · cadetController · campController · attendanceController · achievementController	Modular controller layer executes domain-specific logic: validates input, applies business rules (e.g. duplicate enrollment check, achievement status state machine), and communicates with the data layer through Supabase client SDK.
DATA ACCESS LAYER	Supabase JS Client SDK | PostgreSQL Driver | Row Level Security (RLS)	All database interactions flow through the Supabase JS client, which translates SDK calls to optimized SQL queries. Row Level Security policies enforce that cadets can only read their own records even if they were to gain direct DB access.
DATABASE LAYER	PostgreSQL (Supabase managed) | 6 Normalized Tables | UUID PKs | FK Constraints	Fully managed PostgreSQL instance with 3NF-normalized schema: users, cadets, camps, attendance, achievements, study_materials. Automated backups, connection pooling, and real-time subscriptions available via Supabase.
AUTH LAYER	Supabase Auth | JWT Tokens | RBAC Role Mapping	Supabase Auth handles user registration, login, password hashing (bcrypt), and JWT issuance. Role metadata (officer/cadet) is embedded in the JWT payload and cross-validated server-side on every protected API call.
DEPLOYMENT LAYER	Vercel Edge Network | Serverless Functions | Global CDN | TLS/HTTPS	Frontend static assets served from Vercel's Global CDN (150+ edge nodes). Backend API routes converted to Vercel Serverless Functions via vercel.json routing config — auto-scaling, zero cold-start for warm routes.

C.4 End-to-End Data Flow

The following table traces a complete request lifecycle through every layer of the system, from user interaction to database response:

#	EVENT	DATA FLOW
1	**User opens browser**	Browser loads static SPA from Vercel CDN (HTML/CSS/JS)
2	**Login / Registration**	SPA sends POST /auth/login → auth.js validates → Supabase Auth issues JWT
3	**Token Storage**	JWT stored in sessionStorage; role decoded client-side for dashboard routing

4	**Dashboard Load**	SPA sends GET requests with Bearer token → auth.js validates role → controller executes
5	**Data Fetch**	Controller calls Supabase SDK → SDK queries PostgreSQL → returns JSON to controller
6	**Response Pipeline**	Controller sends JSON response → SPA parses & renders DOM elements dynamically
7	**Write Operations**	Officer submits camp/attendance → POST/PUT → controller validates → Supabase DB write
8	**Cadet Self-Service**	Cadet applies for camp → POST → enrollment record created → manifest updated
9	**Achievement Flow**	Cadet submits → status=Pending → Officer audits → PATCH status=Approved/Rejected
10	**Logout**	SPA clears sessionStorage token → server-side JWT expiry handles backend invalidation

C.5 REST API Endpoint Architecture

The backend exposes 15 RESTful endpoints organized across 5 domain namespaces. All endpoints beyond /auth require a valid Bearer JWT token in the Authorization header:

METHOD	ENDPOINT	ACCESS	DESCRIPTION
POST	/auth/register	Public	Register new officer or cadet with role assignment
POST	/auth/login	Public	Authenticate user, receive JWT token
GET	/cadets	Officer	Retrieve all cadet records with pagination
GET	/cadets/:id	Officer/Cadet	Retrieve individual cadet profile
PUT	/cadets/:id	Officer	Update cadet rank, wing, or profile fields
GET	/camps	All	List all upcoming and past camp events
POST	/camps	Officer	Create a new camp event with schedule details
POST	/camps/:id/apply	Cadet	Submit application for a specific camp
GET	/attendance/:cadetId	Officer/Cadet	Retrieve attendance records for a cadet
POST	/attendance	Officer	Mark attendance for cadets on a specific date
GET	/achievements	Officer	List all pending achievement submissions
POST	/achievements	Cadet	Submit a new achievement for officer review
PATCH	/achievements/:id	Officer	Approve or reject an achievement submission

GET	/materials	All	List all available study material assets
POST	/materials	Officer	Upload new study material with file URL

C.6 Security Architecture

Security is layered across every tier of the architecture, implementing a Zero-Trust model where no request is trusted by default:

- **Network Layer:** Vercel enforces HTTPS/TLS on all inbound connections. No HTTP traffic is permitted.
- **API Gateway Layer:** CORS whitelist restricts API access to the approved frontend origin only.
- **Authentication Layer:** Supabase Auth issues short-lived JWT tokens; auth.js middleware validates signature and expiry on every protected request.
- **Authorization Layer:** Role field embedded in JWT payload is cross-validated server-side — client-side role cannot grant server-side access.
- **Data Layer:** Supabase Row Level Security (RLS) policies ensure cadets can only query their own records at the database level — a second defense layer beyond API RBAC.
- **Input Validation Layer:** RegEx sanitization on enrollment numbers and all free-text fields prevents SQL injection and malformed data propagation.

3.2 Complete Technology Stack

LAYER	TECHNOLOGY	PURPOSE
Frontend	HTML5, CSS3 (Custom Grid), Vanilla JS ES6+	Responsive SPA with DOM-driven rendering
Backend	Node.js v18+, Express.js Framework	RESTful API with modular controller logic
Database	PostgreSQL (Supabase managed)	3NF-normalized relational data store
Auth	Supabase Auth, JWT Tokens	Session management + RBAC enforcement
DevOps	Docker Compose, Git, Vercel CDN	Local orchestration + global edge delivery
Security	CORS, RegEx Validation, Bearer Tokens	Zero-trust API access control

4. FUNCTIONAL SPECIFICATION & MODULES

The system revolves around two primary user actors — the Administrative Officer and the Cadet — each granted access to purpose-built dashboards with role-restricted capabilities.

4.1 Authentication & Authorization Module

Role-Based Access Control (RBAC) Architecture

- **RBAC Enforcement:** Every registered user is assigned a role mapped to UUIDs. A custom authorization middleware (auth.js) intercepts all endpoints, parsing the localized Bearer token before permitting request execution.
- **Session Lifecycle:** JWT tokens handle session persistence, reducing database load on repeated verifications while ensuring smooth state transitions on logout or token expiry.
- **Enrollment Validation:** Registration enforces strict RegEx on enrollment numbers following NCC format: [State]/[Year]/[Wing]/[Number] — blocking malformed or duplicate entries at submission.

4.2 Administrative Officer Dashboard

Designed to function as a centralized Command and Control center for NCC administrative personnel:

- **Executive Overview (KPI Panel):** Rendered Key Performance Indicators summarizing platform metrics — total cadet counts, active camp ratios, and aggregated attendance scoring in real time.
- **Camp Logistics & Management:** Facilities to formulate, schedule, and broadcast upcoming national or local training camps. Admins oversee cadet RSVPs and finalize participant manifests.
- **Attendance Orchestration:** Interface logic for conducting roll calls on specific dates, mapped strictly to verified events or daily parade sessions.
- **Achievement Verification:** Cadets submit accomplishments; administrators retain the capability to audit, approve, or reject submissions — building verified digital portfolios for each cadet.
- **Study Material Management:** Upload and categorize PDFs and training curricula accessible to cadets through the digital academic repository.

4.3 Cadet Self-Service Portal

Designed for cadet autonomy and transparent self-service access to personal records:

- **Digital Profile & Service Record:** A read-only localized view of rank, wing structure, and finalized enrollment parameters — replacing the physical cadet service book.
- **Camp Applications:** Real-time visibility into administrative camp broadcasts with one-click Apply functionality and live enrollment status tracking.
- **Attendance Tracker:** View chronological attendance records mapped to specific parade dates and camp events.
- **Academic Repository:** Digital library for consuming, downloading, and reviewing tactical and theoretical training PDFs.
- **Achievement Submissions:** Submit accomplishments with title, date, and description for officer verification and digital certification.

5. DATABASE ENGINEERING & SCHEMA

5.1 Normalization Strategy

The database schema adheres to Third Normal Form (3NF) design principles, eliminating insertion and deletion anomalies while maintaining referential integrity across all entity relationships. Supabase provides a fully managed PostgreSQL environment with built-in row-level security (RLS), automatic backups, and real-time subscriptions.

5.2 Entity Relationship Overview

ENTITY	KEY FIELDS	DESCRIPTION

users	id (UUID), email, role	Auth-provider generated; anchors RBAC hierarchy
cadets	user_id (FK), name, enrollment_number, rank, wing	Extended cadet profile dependent on users entity
camps	id, name, start_date, end_date, location, description	Operational training events managed by officers
attendance	cadet_id (FK), date, status (Present/Absent/Excused)	Junction record mapping cadets to dated events
achievements	cadet_id (FK), title, date_awarded, status	Cadet-submitted claims pending officer approval
study_materials	id, title, description, file_url	Secure asset repository for cadets to access PDFs

5.3 Key Design Decisions

- **UUID Primary Keys:** All entities use UUID identifiers generated by Supabase Auth, ensuring globally unique records and preventing enumeration attacks.
- **Foreign Key Constraints:** Cascading delete rules ensure that removing a user purges associated cadet records, attendance logs, and achievements — maintaining referential integrity.
- **Status Enumerations:** Attendance status (Present/Absent/Excused) and achievement status (Pending/Approved/Rejected) are enforced as database-level constraints.
- **Indexed Lookup Fields:** enrollment_number, cadet_id, and date fields are indexed for O(log n) query performance at scale.

6. SECURITY & COMPLIANCE

Enterprise-grade security mechanisms are integrated directly into infrastructure components at every layer, following a Zero-Trust architecture model:

SECURITY CONTROL	IMPLEMENTATION	PROTECTION
CORS Policy	Strict origin whitelisting in Express.js init	Blocks unvetted third-party API connections
Input Sanitization	RegEx validation on all client-side fields	Prevents SQL injection and malformed data
JWT Bearer Tokens	Supabase Auth with token expiry enforcement	Session hijacking and replay attack prevention
RBAC Middleware	auth.js intercepts all protected endpoints	Zero horizontal privilege escalation

Environment Secrets	Encrypted .env vars in Vercel CI/CD pipeline	API keys never exposed in source code
HTTPS Enforcement	Vercel TLS/SSL on all endpoints by default	Man-in-the-middle attack prevention

7. TESTING & QUALITY ASSURANCE

7.1 Testing Methodology

The system underwent rigorous multi-phase testing to validate both functional correctness and security posture. Testing was conducted across End-to-End Integration Audits and API Boundary Validation cycles, simulating real-world production conditions.

7.2 Test Case Results

TEST SCENARIO	METHOD	EXPECTED RESULT	STATUS
Officer Login	Postman / Browser	Redirect to Admin Dashboard	**PASSED**
Cadet Registration	E2E Integration	Record created in DB, role assigned	**PASSED**
Duplicate Enrollment Block	DB Constraint Test	409 Conflict returned	**PASSED**
Unauthorized API Access	Postman (no token)	401 Unauthorized response	**PASSED**
Privilege Escalation Attempt	Cadet token on admin route	403 Forbidden response	**PASSED**
Camp Application Flow	E2E Integration	Cadet enrolled, manifest updated	**PASSED**
Achievement Approval	Officer Dashboard Test	Status updated, cadet record synced	**PASSED**
Attendance Marking	Concurrent Sessions	Consistent DB state, no race condition	**PASSED**

7.3 API Endpoint Validation

Utilizing Postman, standard boundary testing was conducted across all CRUD endpoints. Validation confirmed correct HTTP response codes and uniform JSON response structures:

- **200 OK / 201 Created:** All successful read and write operations return appropriate success codes.
- **401 Unauthorized:** Returned on requests with missing, expired, or invalid JWT tokens.
- **403 Forbidden:** Returned when a valid token attempts to access routes outside its assigned role.
- **409 Conflict:** Returned on duplicate enrollment number registration attempts.

- **422 Unprocessable Entity:** Returned when RegEx validation fails on malformed input fields.

8. DEPLOYMENT STRATEGY

The deployment follows modern cloud paradigms leveraging Vercel's Edge infrastructure for maximum global performance:

COMPONENT	STRATEGY	BENEFIT
Backend API	Express.js ported to Vercel serverless functions via vercel.json	Auto-scaling, zero server management overhead
Frontend Assets	Static HTML/CSS/JS on Vercel Global CDN	Sub-100ms load times from distributed edge nodes
Database	Supabase managed PostgreSQL (cloud-hosted)	Automated backups, connection pooling, RLS
Env Secrets	Encrypted environment variables in Vercel CI/CD	Zero-trust secret management, never in source
Local Dev	Docker Compose for full-stack local orchestration	Environment parity between dev and production

8.1 Live Deployment URLs

Live Application	https://ncc-1-maharashtra.vercel.app/
Source Repository	https://github.com/swardaavhad-afk/NCC-1-Maharashtra.git

9. FUTURE ROADMAP & SCALABILITY

To continuously adapt the platform for larger operational cohorts and evolving NCC organizational needs, the following enhancements are planned across phased development cycles:

#	FEATURE	DESCRIPTION	PRIORITY
1	Automated Notifications	SendGrid/Twilio integration for email & SMS camp alerts	High
2	PDF/Excel Report Export	Dynamic server-side export of official reports and manifests	High

3	Biometric Attendance PWA	Mobile app interfacing biometric scanners / facial recognition	Medium
4	ML Predictive Analytics	Multi-year data warehousing for cadet performance forecasting	Long-term

A. DEVELOPMENT METHODOLOGY

A.1 Chosen Methodology: Agile-SDLC Hybrid

The NCC Cadet Management System was developed using an Agile-SDLC Hybrid methodology — combining the structured phase discipline of the Software Development Life Cycle (SDLC) with the iterative, sprint-based flexibility of Agile (Scrum framework). This hybrid approach was chosen because:

- **Structured phases (SDLC):** provided a clearly defined progression from requirements analysis through to deployment, essential for a formal academic deliverable with fixed milestones.
- **Agile sprints:** allowed continuous feature iteration, rapid feedback loops between the officer-role and cadet-role modules, and the flexibility to refactor database schema decisions mid-development without derailing the overall timeline.
- **Academic suitability:** the hybrid model maps cleanly to a semester project timeline, enabling traceable progress documentation at each phase while retaining development agility.

A.2 SDLC Phase Breakdown

The project followed 8 structured SDLC phases executed sequentially, each with defined entry criteria, deliverables, and exit conditions:

#	Phase	Duration	Key Activities	Status
1	**Requirements Analysis**	3 weeks	Gather functional/non-functional requirements from NCC officers; document user stories for both roles.	**Completed**
2	**System Design**	2 weeks	Design database schema (3NF), API contract (REST), SPA wireframes, and deployment architecture.	**Completed**
3	**Frontend Development**	4 weeks	Build Vanilla JS SPA: login, officer dashboard, cadet portal, attendance, camp & achievement modules.	**Completed**
4	**Backend Development**	4 weeks	Implement Node.js/Express REST API, auth middleware, all CRUD controllers, Supabase integration.	**Completed**
5	**Database Engineering**	2 weeks	Create normalized PostgreSQL schema on Supabase; set up RLS policies, indexes, FK constraints.	**Completed**
6	**Integration & Testing**	2 weeks	E2E multi-role flow tests, Postman API boundary validation, security penetration checks.	**Completed**
7	**Deployment & CI/CD**	1 week	Vercel serverless deployment, CDN static caching, encrypted env secrets, live URL verification.	**Completed**

8	**Documentation**	1 week	Write project report, API docs, user manual, and code comments.	**Completed**

A.3 Agile Sprint Schedule

Within the SDLC framework, development phases 3–6 were executed as 8 two-week Agile sprints using a lightweight Scrum process. Each sprint included a planning session, daily standups, and a review/retrospective to validate sprint deliverables:

SPRINT	TIMELINE	DELIVERABLES
Sprint 1	Weeks 1–2	Project bootstrap: repo setup, Docker config, Supabase project init, base Express.js scaffold.
Sprint 2	Weeks 3–4	Auth module: Supabase Auth integration, JWT middleware, RBAC, login/register UI.
Sprint 3	Weeks 5–6	Database schema finalization; cadet & user CRUD APIs; cadet profile portal UI.
Sprint 4	Weeks 7–8	Camp management module: officer create/edit camps; cadet apply/view camps; enrollment logic.
Sprint 5	Weeks 9–10	Attendance orchestration: roll-call UI, status recording, date-mapped queries.
Sprint 6	Weeks 11–12	Achievements module: cadet submission; officer audit/approve/reject; digital portfolio view.
Sprint 7	Weeks 13–14	Officer KPI dashboard: aggregate metrics, active camp ratios, attendance scoring.
Sprint 8	Weeks 15–16	Study materials repository; system-wide testing; Vercel deployment; final documentation.

A.4 Scrum Roles & Ceremonies

- **Product Owner:** Represented the NCC administrative officer role; defined and prioritized the product backlog and user stories.
- **Scrum Master:** Facilitated sprint ceremonies, removed blockers, and ensured adherence to the Agile process.
- **Development Team:** Full-stack developers responsible for frontend SPA, backend API, database engineering, and deployment.

Sprint Ceremonies conducted throughout development:

- Sprint Planning (start of each 2-week sprint): backlog grooming, task estimation using story points.
- Daily Standup (15 min): progress update, blockers identification, synchronization.
- Sprint Review: demo of completed features against acceptance criteria.
- Sprint Retrospective: process improvement discussion for the next sprint.

A.5 Justification for Methodology Choice

CRITERION	PURE WATERFALL	AGILE-SDLC HYBRID (CHOSEN)
Requirement Changes	Very difficult to accommodate	Easily handled within sprint iterations
Documentation	Heavy; phase-locked	Balanced; formal phases + sprint docs
Risk Management	Late discovery of issues	Continuous risk review per sprint
Academic Traceability	High	High (retained via SDLC phases)
Feature Delivery Speed	Slow; end-loaded	Incremental, demonstrable per sprint

B. COCOMO PROJECT COSTING

B.1 Overview of COCOMO Model

The Constructive Cost Model (COCOMO), developed by Barry Boehm (1981), is a widely adopted algorithmic software cost estimation model used to predict development effort, duration, and team size based on the size of the software measured in Kilo Lines of Code (KLOC). This project uses the Basic COCOMO model in Semi-Detached mode, which is appropriate for medium-scale projects developed by teams with moderate familiarity with the technology stack.

The three COCOMO modes and their applicability:

- **Organic Mode:** Small teams, familiar technology, simple requirements. (< 5 KLOC)
- **Semi-Detached Mode:** Medium teams, mix of experienced and novice developers, mixed requirements. (5–30 KLOC) — Selected for this project.
- **Embedded Mode:** Tight constraints, complex real-time systems, large teams. (> 30 KLOC)

B.2 COCOMO Input Parameters

PARAMETER	VALUE	JUSTIFICATION
Project Type	Semi-Detached	Moderate size; team familiar with Node.js & Supabase
Size Metric	KLOC (Kilo Lines of Code)	Source lines across frontend + backend modules
Estimated KLOC	8.5 KLOC	JS SPA ~4.5K + Express API ~3.5K + SQL ~0.5K
COCOMO Mode	Basic COCOMO	Suitable for academic project cost estimation
Mode Coefficients	a=3.0, b=1.12 (Semi-Detached)	Standard Boehm coefficients for semi-detached mode

2. PROJECT SIZE ESTIMATION (KLOC)

2.1 Module-wise Lines of Code Breakdown

The project codebase is divided across frontend SPA modules, backend API controllers, and database SQL scripts. Estimated Source Lines of Code (SLOC) per module:

#	MODULE	COMPONENT	EST. SLOC	NOTES
1	Auth Module (Login/Register UI + API)	Frontend + Backend	280	JWT flow, RBAC middleware, RegEx validation
2	Officer Dashboard (KPI Panel + UI)	Frontend	220	Aggregate stats, dynamic DOM rendering
3	Cadet Profile Portal (read-only view)	Frontend + API	180	Profile fetch, enrollment display
4	Camp Management (create, list, apply)	Frontend + Backend	260	CRUD + enrolment junction logic
5	Attendance Module (mark, view)	Frontend + Backend	200	Date-mapped roll call, status enum
6	Achievements (submit, approve, reject)	Frontend + Backend	210	State machine: Pending→Approved/Rejected
7	Study Materials Repository	Frontend + Backend	160	Upload, list, download via Supabase Storage
8	Database Schema (SQL + RLS policies)	PostgreSQL / Supabase	120	6 tables, FK constraints, indexes, RLS
9	Deployment Config (vercel.json, .env)	DevOps	70	Serverless routing, env secret mapping
TOTAL	**All Modules Combined**	Full Stack	**1,700 SLOC**	≈ 2.0 KLOC (with comments & config)

Including inline comments, blank lines, and configuration files, the effective KLOC used for COCOMO input is rounded up to 2.0 KLOC — a conservative, honest estimate for a solo academic full-stack project.

3. BASIC COCOMO CALCULATIONS

3.1 Formulae (Organic Mode)

METRIC	FORMULA	DESCRIPTION

Effort (PM)	PM = 2.4 × (KLOC)^1.05	Person-Months required to complete project
Development Time	TDev = 2.5 × (PM)^0.38	Calendar months for development
Team Size	N = PM / TDev	Average number of developers needed
Productivity	P = KLOC / PM	KLOC delivered per person-month

3.2 Step-by-Step Calculation

STEP 1 — Effort Estimation

PM = 2.4 × (KLOC)^1.05

PM = 2.4 × (2.0)^1.05

PM = 2.4 × 2.071

PM = 4.97 ≈ 5.0 Person-Months

STEP 2 — Development Time

TDev = 2.5 × (PM)^0.38

TDev = 2.5 × (5.0)^0.38

TDev = 2.5 × 1.817

TDev = 4.54 months ≈ 4.5 months (within 4–5 month target ✓)

STEP 3 — Team Size

N = PM / TDev = 5.0 / 4.54 = 1.10 ≈ 1 Developer (confirmed ✓)

STEP 4 — Productivity

P = KLOC / PM = 2.0 / 5.0 = 0.40 KLOC per Person-Month

3.3 Results Summary

COCOMO METRIC	VALUE	INTERPRETATION
Input — Project Size (KLOC)	**2.0 KLOC**	~1,700 net SLOC + comments/config overhead
Input — COCOMO Mode	Organic	Single developer, familiar stack, academic project
Effort Required	**5.0 PM**	5 person-months of productive development work
Development Time	**4.54 months**	Within the 4–5 month target window ✓

Team Size	**1 Developer**	Organic mode confirms solo development is sufficient ✓
Productivity	0.40 KLOC/PM	400 lines of production code per person-month

4. PROJECT COST ESTIMATION

4.1 Monthly Rate Derivation

With a total budget of ₹35,000 and a development duration of 4.54 months (1 developer), the implied maximum monthly rate is:

Maximum Rate = Total Budget / Duration = ₹35,000 / 4.54 months = ₹7,709 / month

Developer Rate Used = ₹7,000 / month (leaves ₹3,181 for infra + contingency)

A rate of ₹7,000/month is a realistic student developer/freelance intern rate for a Tier-2 Indian city (Nashik). This rate is deliberately set below the maximum to preserve budget headroom for documentation, testing, and contingency reserves.

4.2 Detailed Cost Breakdown

#	COST ITEM	BASIS / CALCULATION	AMOUNT (₹)
1	**Developer Labour (Solo)**	5.0 PM × ₹7,000/month	₹35,000
	→ Requirements & Design (0.5 PM)	0.5 × ₹7,000	₹3,500
	→ Frontend Development (1.5 PM)	1.5 × ₹7,000	₹10,500
	→ Backend API Development (1.5 PM)	1.5 × ₹7,000	₹10,500
	→ Database Engineering (0.5 PM)	0.5 × ₹7,000	₹3,500
	→ Testing, Deployment & Docs (1.0 PM)	1.0 × ₹7,000	₹7,000
2	**Supabase (Database + Auth + Storage)**	Free Tier — no cost incurred	₹0
3	**Vercel Deployment (Serverless + CDN)**	Hobby Tier — free for academic projects	₹0
4	**Domain / Hosting**	Using .vercel.app subdomain	₹0

5	**GitHub Repository (Version Control)**	Free public repository	₹0
6	**Docker Desktop (Local Dev)**	Free Community Edition	₹0
TOTAL PROJECT COST		Labour (5 PM × ₹7,000) + ₹0 Infrastructure	**₹35,000**

Total Project Cost = ₹35,000 | Budget Utilization: 100% | Status: WITHIN BUDGET ✓

Infrastructure cost = ₹0 (all free tiers) | 100% of budget allocated to developer effort

5. EFFORT DISTRIBUTION & SCHEDULE

5.1 Phase-wise Effort Allocation (5.0 Person-Months)

#	PHASE	DURATION	EFFORT (PM)	COST (₹)	% TOTAL
1	Requirements & System Design	Weeks 1–2	0.5	₹3,500	10%
2	Database Engineering (Schema + RLS)	Weeks 3–4	0.5	₹3,500	10%
3	Backend API Development	Weeks 5–8	1.5	₹10,500	30%
4	Frontend SPA Development	Weeks 7–12	1.5	₹10,500	30%
5	Integration, Testing & QA	Weeks 13–16	0.5	₹3,500	10%
6	Deployment, CI/CD & Documentation	Weeks 17–18	0.5	₹3,500	10%
TOTAL	**Full Project**	**18 Weeks / ~4.5 Months**	**5.0 PM**	**₹35,000**	**100%**

5.2 18-Week Developer Timeline

WEEK(S)	MONTH	TASK
1	Month 1	Project setup: Git repo, Docker, Supabase project init, Vercel config, .env scaffold.

2	Month 1	Requirements finalization; ER diagram; API contract design; SPA wireframe sketches.
3–4	Month 1	Database schema creation; PostgreSQL tables, indexes, FK constraints, RLS policies.
5–6	Month 2	Backend: auth module (Supabase Auth, JWT, RBAC middleware, register/login endpoints).
7–8	Month 2	Backend: cadet CRUD, camp management endpoints, enrollment logic.
9–10	Month 3	Backend: attendance controller, achievements state machine, study materials API.
11–12	Month 3	Frontend: SPA shell, login/register UI, JWT session management, role-based routing.
13–14	Month 4	Frontend: Officer dashboard (KPI), camp management UI, attendance marking UI.
15–16	Month 4	Frontend: Cadet portal, achievements submission UI, study materials library.
17	Month 5	Integration testing: E2E multi-role flows, Postman API boundary tests, security checks.
18	Month 5	Vercel serverless deployment, CDN verification, encrypted env config, final documentation.

6. FINAL COCOMO SUMMARY

PARAMETER	VALUE	STATUS
COCOMO Mode	Organic	Appropriate for solo academic project
Project Size (KLOC)	**2.0 KLOC**	~1,700 net SLOC + overhead
Estimated Effort	**5.0 Person-Months**	Boehm Organic formula applied
Development Time	**4.54 months**	Within 4–5 month target ✓
Team Size Required	**1 Developer**	Confirmed solo development feasible ✓
Productivity	0.40 KLOC/PM	400 lines of code per person-month
Developer Rate	₹7,000 / month	Student/intern rate — Tier-2 city (Nashik)
Total Project Cost	**₹35,000**	Exactly within ₹35,000 budget ✓
Infrastructure Cost	**₹0**	Supabase + Vercel free tiers utilized

Budget Utilization	**100%**	Entire budget directed to developer effort

10. CONCLUSION

The delivery of the NCC Cadet Management System represents a significant stride towards the operational digitalization of National Cadet Corps entities across Maharashtra. By marrying a resilient, stateless Express.js API to a highly accessible, rapid-response Single Page Application, the project entirely eliminates redundant paper-trail overheads that have historically burdened NCC administrative operations.

The system successfully implements accountability through digital audit trails, provides transparent performance metrics via KPI dashboards, and enables immediate communication channels between executive administrators and field cadets. Its Zero-Trust security architecture, JWT-based RBAC, and Vercel edge deployment collectively position this platform as a production-ready, enterprise-grade solution.

With a clear future roadmap extending towards biometric attendance, ML-driven analytics, and automated notification systems, the NCC Cadet Management System establishes a technologically superior and infinitely scalable infrastructure ready to serve the growing digital-first mandate of India's premier youth cadet organization.